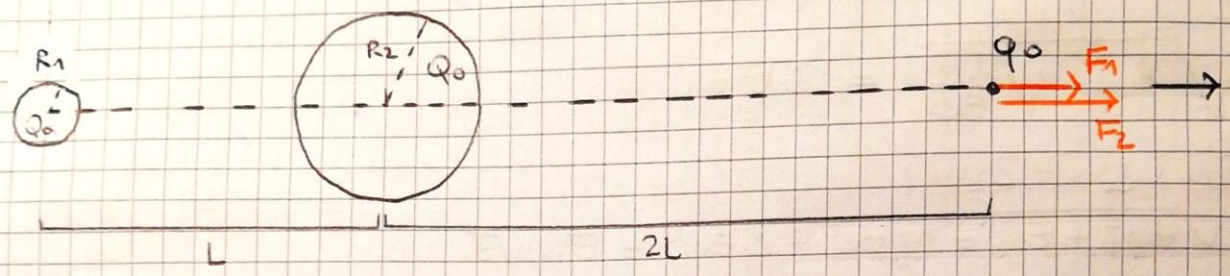
APPELLO 8/07/2016

ESERCIZIO 1



a) $\underline{F} = \underline{F}_1 + \underline{F}_2 = \frac{13}{36} \frac{Q_0 q_0}{4\pi\varepsilon_0 L^2} \underline{u}_x$

b) $L = -\Delta U = U_i - \cancel{U_f}^{\,0} = U_i = U_{12} + U_{22} =$

$= \frac{q_0 Q_0}{4\pi\varepsilon_0 L} + \frac{q_0 Q_0}{4\pi\varepsilon_0 3L} = \frac{5 q_0 Q_0}{24\pi\varepsilon_0 L}$

c) $2Q_0 = Q_1 + Q_2$

$V_1 = V_2 \Rightarrow \frac{Q_1}{\cancel{4\pi\varepsilon_0} R_1} = \frac{Q_2}{\cancel{4\pi\varepsilon_0} R_2}$

$\Rightarrow 2Q_0 = Q_2\left(\frac{R_1}{R_2} + 1\right) = Q_2\left(\frac{R_1 + R_2}{R_2}\right)$

$\Rightarrow Q_2 = \frac{2Q_0 R_2}{R_1 + R_2}$

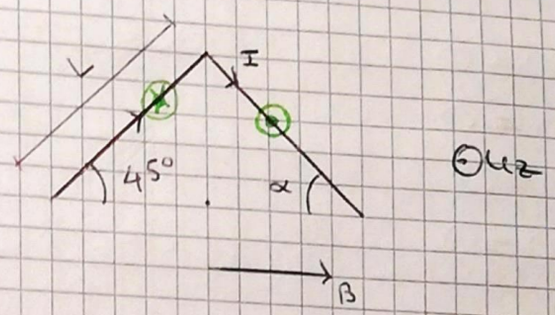
$Q_1 = 2Q_0 - Q_2 = 2Q_0\left(1 - \frac{R_2}{R_1 + R_2}\right) = \frac{2Q_0 R_1}{R_1 + R_2}$

ESERCIZIO 2

$\underline{F}_1 = -BIL \operatorname{sen}\alpha \; \underline{u}_z$

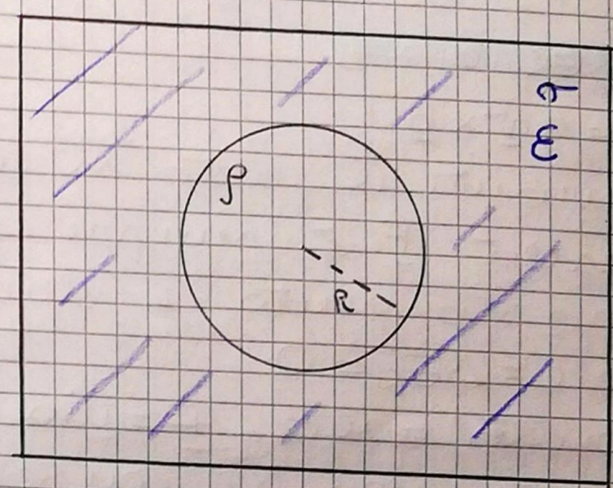
$\underline{F}_2 = BIL \cdot \operatorname{sen}\alpha \; \underline{u}_z$

$F_{TOT} = 0$



ESERCIZIO 3

$\rho = \rho_0 e^{-\frac{\sigma}{\varepsilon}t}$



All'interno della sfera si ha

$\underline{E} = \frac{\rho}{3\varepsilon} r \, \underline{u}_r$

fuori

$\underline{E} = \frac{\rho R^3}{3\varepsilon r^2} \underline{u}_r$

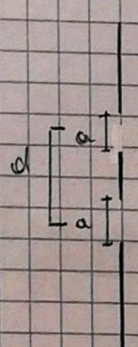
$$\underline{J} = \begin{cases} \frac{\sigma \rho_0 r}{3\varepsilon} e^{-\frac{\sigma}{\varepsilon}t} \underline{u}_r & r<R \\ \frac{\sigma \rho_0 R^3}{3\varepsilon r^2} e^{-\frac{\sigma}{\varepsilon}t} \underline{u}_r & R \leq r \end{cases}$$

$\underline{J}_s = \varepsilon \frac{\partial \underline{E}}{\partial t}$

ESERCIZIO 4

~~I max coincidono se~~

Il max del 4° ordine manca se esso coincide con un minimo di diffrazione. dunque



$\operatorname{sen}\theta_4^{MAX,INT} = 4\frac{\lambda}{d}$

$\parallel$

$\operatorname{sen}\theta_{n_d}^{MINIDIFF} = n_d \frac{\lambda}{a} \Rightarrow a = n_d \frac{d}{4}$

Ma $a < d \Rightarrow n_d < 4$

$\Rightarrow a = n_d \frac{d}{4}$ con $n_d = 1, 2, 3$